FUNDAMENTALS OF CYBER SECURITY
MODULE 4 — HANDWRITTEN QUESTIONS & ANSWERS

Exam-ready notes prepared from the 11 topics listed under Module 4 in the uploaded FCSDF Question Bank.

Q1. Explain Cybercrime Scene Investigation Process with the help of diagram.

Answer:

Cybercrime Scene Investigation is the systematic process of identifying, collecting, preserving, examining and documenting digital evidence related to a cybercrime.

Steps:

1. Preparation: Obtain legal authorization, prepare tools and plan the investigation.
2. Securing the scene: Restrict access and prevent anyone from altering devices or evidence.
3. Identification: Identify computers, mobiles, storage devices, networks and possible evidence.
4. Documentation: Photograph the scene, record device condition, connections, date, time and people present.
5. Collection: Collect relevant digital devices and data using approved procedures.
6. Preservation: Protect evidence from alteration. Use write blockers and forensic imaging where appropriate.
7. Examination: Extract relevant files, logs, metadata and other digital artefacts.
8. Analysis: Interpret evidence and reconstruct events or user activity.
9. Reporting: Prepare a clear report of methods, findings and conclusions.
10. Presentation: Present findings to authorized investigators or court.

Flow diagram:

Preparation

↓

Secure the Scene

↓

Identify & Document Evidence

↓
Collect Evidence
↓
Preserve Evidence
↓
Examine & Analyse
↓
Prepare Report
↓
Present Findings

Conclusion: A proper investigation protects the integrity of digital evidence and helps establish what happened during a cybercrime.

Q2. Describe process of Identifying and Documenting the Scene & Evidence of Cyber Crime Scene with an example.

Answer:

Identification and documentation are important steps in cybercrime investigation. They help locate possible evidence and create a reliable record of the scene before devices are handled.

Identification process:

1. Secure the area and prevent unauthorized access.
2. Identify computers, laptops, mobile phones, servers and storage media.
3. Identify network devices such as routers, switches and external drives.
4. Locate possible evidence such as emails, files, logs, browser history and cloud accounts.
5. Note devices that are powered on, powered off or connected to a network.
6. Identify people who may have knowledge of the system.

Documentation process:

- Photograph the complete scene and each device.
- Record date, time, location and investigator details.
- Record make, model, serial number and condition of each device.
- Draw diagrams showing cable and network connections.
- Label each item with a unique evidence ID.
- Record collection method, packaging and transfer details.

- Maintain a chain of custody record.

Example:

In an online banking fraud case, investigators find a laptop, mobile phone and router. They photograph the setup, record serial numbers, note the device status, label each item and document who collected it. The devices are then preserved for forensic examination.

Conclusion: Proper identification and documentation prevent confusion, support chain of custody and make evidence more reliable.

Q3. Demonstrate Chain of Custody form with an example.

Answer:

Chain of Custody is the chronological written record of the collection, possession, transfer, storage and handling of digital evidence. It proves who handled the evidence and helps demonstrate that it was not improperly altered.

Main details in a chain of custody form:

1. Case number and case title.
2. Evidence identification number.
3. Description of the evidence.
4. Date, time and place of collection.
5. Name and signature of collecting officer.
6. Evidence seal or package number.
7. Details of every transfer.
8. Purpose of transfer and receiving person's signature.
9. Storage location and condition of seal.

Sample form:

Case No.: CC-2026-014
Evidence ID: E-01
Description: Laptop, make/model and serial number recorded
Collected from: Suspected cybercrime scene
Collected by: Investigator A
Date and time: 31-10-2026, 10:30 AM
Seal No.: SEAL-101

Transfer record:

From: Investigator A

To: Forensic Examiner B
Date/time: 31-10-2026, 12:00 PM
Purpose: Forensic imaging and examination
Signatures: Both parties sign the transfer record.

Conclusion: Every transfer must be recorded immediately. A complete chain of custody supports authenticity and admissibility of digital evidence.

Q4. Explain Digital Forensic Lifecycle with neat labelled diagram.

Answer:
Digital Forensic Lifecycle is a sequence of stages followed to identify, preserve, collect, examine, analyse and report digital evidence.

Stages:
1. Identification: Locate possible sources of digital evidence.
2. Preservation: Protect evidence from modification or destruction.
3. Collection / Acquisition: Collect devices or create forensic copies using approved methods.
4. Examination: Extract and organize relevant information from acquired evidence.
5. Analysis: Interpret the extracted data and connect events.
6. Documentation: Record tools, methods, dates, actions and observations.
7. Reporting / Presentation: Communicate findings clearly to investigators or court.

Diagram:
Identification
↓
Preservation
↓
Collection / Acquisition
↓
Examination
↓
Analysis
↓

Documentation

↓

Reporting / Presentation

Note: The process may be iterative. New findings can lead investigators to identify and examine additional evidence.

Conclusion: The digital forensic lifecycle provides a structured and repeatable method for handling digital evidence while maintaining its integrity.

Q5. Explain challenges and applications of Digital Forensics.

Answer:

Digital Forensics is the process of identifying, collecting, preserving, examining and analysing digital evidence from electronic devices.

Challenges:

1. Large data volume: Modern devices store huge amounts of information.
2. Encryption: Encrypted files may be difficult to access without valid keys.
3. Anti-forensics: Attackers may delete, hide or alter evidence.
4. Rapid technology changes: New devices, operating systems and cloud services require updated tools.
5. Volatile data: Information in RAM and active network connections may disappear when a device is powered off.
6. Legal and privacy issues: Investigators must follow laws, warrants and privacy requirements.
7. Evidence integrity: Evidence must be protected from accidental modification.
8. Cross-border data: Data may be stored in different countries and jurisdictions.

Applications:

- Criminal investigations involving hacking, fraud and identity theft.
- Corporate investigations of data leaks and insider threats.
- Incident response and malware investigations.
- Recovery and examination of deleted files.
- Mobile phone and social media investigations.
- Intellectual property theft investigations.

- Supporting civil and court cases.

Conclusion: Digital forensics faces technical and legal challenges but plays an important role in investigating cyber incidents and presenting reliable evidence.

Q6. Explain Chain of Custody Process.

Answer:

Chain of Custody is the documented history of evidence from the time it is collected until it is presented, transferred, stored or disposed of according to procedure.

Process:

1. Identification: Assign a unique evidence ID to every item.
2. Collection: Record where, when and by whom the evidence was collected.
3. Packaging: Place evidence in suitable protective packaging and apply a tamper-evident seal.
4. Labelling: Mention case number, evidence ID, description and collection details.
5. Transfer: Record the name of the person handing over and receiving the evidence.
6. Storage: Keep evidence in a secure, access-controlled location.
7. Examination: Record every access, action, tool and relevant result.
8. Return or disposal: Record the final authorized transfer or disposal.

Important rules:

- Maintain a chronological record.
- Never leave evidence unattended.
- Use tamper-evident packaging.
- Restrict access to authorized persons.
- Record date, time and signatures for each transfer.
- Use cryptographic hashes to help verify that forensic copies have not changed.

Conclusion: Chain of custody maintains accountability and supports the authenticity and reliability of digital evidence.

Q7. Write a short note on Forensic Imaging.

Answer:

Forensic Imaging is the process of creating a bit-by-bit copy of a digital storage device, such as a hard disk, SSD or USB drive, for investigation.

Purpose:

- Preserve the original evidence.
- Allow examination without directly working on the original device.
- Capture allocated, unallocated and other accessible disk areas.
- Support repeatable forensic analysis.

Steps:

1. Identify and document the original storage device.
2. Connect it using a suitable write blocker where applicable.
3. Select a trusted forensic imaging tool.
4. Create a bit-stream image of the source device.
5. Calculate cryptographic hash values, such as SHA-256, for verification.
6. Verify the image and record tool details, date and time.
7. Securely store the original and work on the verified forensic copy.

Common formats:

- Raw / DD image.
- E01 / Expert Witness Format.

Conclusion: Forensic imaging creates a reliable working copy and helps preserve the original digital evidence for investigation.

Q8. Describe Steps in Cloning.

Answer:

Cloning is the process of creating an exact copy of a storage device or its data onto another storage medium. In digital forensics, a forensic clone may be used for examination while preserving the original.

Steps:

1. Identify the source device and destination device.
2. Record the make, model, serial number and capacity of both devices.
3. Ensure the destination has sufficient capacity.
4. Connect the source using a write blocker where applicable.
5. Select a trusted forensic cloning or imaging tool.
6. Select the correct source and destination carefully.
7. Start the bit-by-bit copying process.

8. Verify the copy using cryptographic hash values where supported.
9. Record the tool, settings, date, time and verification results.
10. Secure the original evidence and label the clone.

Precautions:

- Never accidentally overwrite the original evidence.
- Use a clean destination drive.
- Maintain chain of custody.
- Record errors or unreadable sectors.
- Verify the resulting copy.

Conclusion: Proper cloning helps investigators work with a duplicate while protecting the original evidence.

Q9. Differentiate between Live System Forensics and Dead System Forensics.

Answer:

Live System Forensics is the collection and examination of evidence from a computer while it is running. Dead System Forensics examines a powered-off system or its storage media.

Live System Forensics:

- System state: Computer is powered on.
- Evidence: RAM, running processes, active network connections, logged-in users and temporary data.
- Advantage: Can capture volatile information that may disappear after shutdown.
- Limitation: Actions performed during collection may change system state.
- Example: Collecting active network connections from a running server.

Dead System Forensics:

- System state: Computer is powered off.
- Evidence: Files, disk partitions, deleted data and file-system metadata.
- Advantage: Easier to preserve storage data in a controlled manner.
- Limitation: Volatile information such as RAM contents is lost after power-off.
- Example: Creating a forensic image of a seized hard disk.

Key difference:

Live forensics focuses on volatile and active system data, while dead forensics focuses mainly on non-volatile storage evidence.

Conclusion: Investigators choose the method according to the case. In some situations, live data must be collected before the system is shut down.

Q10. Explain different Techniques and Methods in Preservation of Evidence.

Answer:

Evidence Preservation means protecting digital evidence from alteration, deletion, damage or unauthorized access during an investigation.

Techniques and methods:

1. Secure the scene: Restrict access to devices and prevent unauthorized handling.
2. Photograph and document: Record the condition, connections and screen contents where appropriate.
3. Use write blockers: Prevent changes to original storage media during acquisition.
4. Forensic imaging: Create a bit-by-bit copy for examination.
5. Hash verification: Calculate hashes such as SHA-256 and compare values to detect changes.
6. Chain of custody: Record every person who handles or transfers evidence.
7. Proper packaging: Use suitable anti-static and tamper-evident packaging.
8. Secure storage: Store evidence in access-controlled and monitored locations.
9. Preserve volatile data: When justified, collect RAM and active system information before shutdown.
10. Backup and redundancy: Maintain secure copies of forensic images and records.
11. Time documentation: Record timestamps and relevant time-zone details.
12. Legal compliance: Follow authorization, privacy and evidence-handling procedures.

Conclusion: Evidence preservation maintains integrity and helps ensure that

digital evidence remains reliable throughout the investigation.

Q11. Write a short note on Hashing.

Answer:

Hashing is the process of converting input data of any size into a fixed-length value called a hash or message digest using a mathematical algorithm.

Features:

- Fixed-length output: The digest has a fixed size for a given algorithm.
- Deterministic: The same input produces the same hash.
- One-way: It is designed to be computationally infeasible to recover the original input from the hash.
- Avalanche effect: A small input change produces a significantly different hash.
- Collision resistance: It should be difficult to find two different inputs with the same hash.

Common algorithms:

- MD5: Produces a 128-bit hash; not considered secure for collision-resistant evidence verification.
- SHA-1: Produces a 160-bit hash; deprecated for collision-sensitive security uses.
- SHA-256: Part of SHA-2 and produces a 256-bit hash; commonly used for integrity verification.

Applications:

1. Checking file integrity.
2. Verifying forensic images.
3. Detecting changes in evidence.
4. Password storage when used with dedicated password-hashing schemes and salts.

Conclusion: Hashing is an important method for verifying digital evidence integrity. Modern forensic work should prefer secure algorithms such as SHA-256.

Q12. Describe Process of Using Hashing in Evidence Handling.

Answer:

Hashing is used in digital forensics to verify that evidence or a forensic

copy has not changed during acquisition, transfer or examination.

Process:

1. Identify the original evidence device or file.
2. Record its description and unique evidence ID.
3. Use a trusted forensic tool to calculate a cryptographic hash, preferably SHA-256.
4. Record the hash value, algorithm, date, time and tool used.
5. Create a forensic image or copy using approved procedures.
6. Calculate the hash of the acquired image.
7. Compare the source and image hashes when the acquisition method and format permit a direct comparison.
8. Store the image and hash record securely.
9. Recalculate the hash after transfer or before examination to verify integrity.
10. Include hash values and verification results in the forensic report.

Example:

Original evidence image hash:

SHA-256 = ABC...123

After transfer, the image is hashed again.

If the resulting SHA-256 value matches, it provides strong evidence that the image has not changed.

Note: A physical device and its container image may have different hash values depending on the imaging format and acquisition process. Investigators must compare the correct corresponding data.

Conclusion: Hashing provides a practical way to detect changes and support the integrity of digital evidence.

Q13. Explain Examination of Evidence and tasks involved in it.

Answer:

Examination of Evidence is the process of extracting, organizing and identifying relevant information from collected digital evidence using forensic methods and tools.

Tasks involved:

1. Verify evidence: Check labels, seals, documentation and hash values.
2. Prepare working copy: Use a verified forensic image instead of the

original wherever possible.

3. Identify file systems: Examine partitions, directories and file-system structures.

4. Recover data: Search for deleted, hidden or damaged files where possible.

5. Extract artefacts: Collect browser history, emails, logs, documents and application data.

6. Examine metadata: Review timestamps, file properties and other metadata.

7. Search keywords: Locate terms, names, dates or file types relevant to the case.

8. Document findings: Record tools, versions, methods and extracted information.

Common tools may include Autopsy, The Sleuth Kit, EnCase and FTK, depending on availability and authorization.

Conclusion: Examination extracts relevant digital information and prepares it for deeper analysis.

Q14. Explain Analysis of Evidence and tasks involved in it.

Answer:

Analysis of Evidence is the process of interpreting examined digital data to understand events, identify relationships and determine what may have happened during an incident.

Tasks involved:

1. Correlate evidence: Compare files, logs, emails and device records.

2. Build timelines: Arrange events according to timestamps.

3. Identify user activity: Examine logins, file access and application usage.

4. Analyse communication: Review relevant emails, messages and network records.

5. Investigate malware or attacks: Study suspicious files, processes and system changes.

6. Recover deleted information: Interpret recovered files and file-system artefacts.

7. Establish relationships: Connect accounts, devices, files and events.

8. Validate findings: Cross-check conclusions against multiple sources.

9. Document limitations: Record missing data, time differences and uncertainty.

Example:

A login log shows access at 9:00 PM, a file timestamp shows modification at 9:05 PM, and an email was sent at 9:10 PM. Investigators correlate these records to reconstruct a possible sequence of activity, while checking timestamp reliability.

Conclusion: Examination extracts information; analysis interprets it and develops evidence-based findings.

Q15. Write a short note on Report Generation in Digital Forensics with sample of Report Structure.

Answer:

A Digital Forensic Report is a formal document that explains the investigation, methods used, evidence examined, findings and conclusions.

Purpose:

- Communicates technical findings clearly.
- Creates a permanent investigation record.
- Supports decision-making and legal proceedings.
- Allows another examiner to understand and review the work.

Sample report structure:

1. Case details: Case number, title, date and investigating organization.
2. Objective: Purpose and scope of the examination.
3. Authorization: Legal authority or permission for the investigation.
4. Evidence details: Evidence ID, description and source.
5. Chain of custody: Collection, transfer and storage history.
6. Tools and methods: Software, hardware, versions and procedures used.
7. Examination steps: Actions performed on the evidence.
8. Findings: Relevant files, logs, metadata and recovered data.
9. Analysis: Interpretation and timeline of events.
10. Hash values: Integrity verification details.
11. Limitations: Missing, inaccessible or damaged evidence and other constraints.
12. Conclusion: Summary of findings related to the investigation.

13. Appendices: Screenshots, tables and supporting records.
14. Examiner details: Name, role, signature and report date.

Conclusion: A clear, factual and reproducible report helps investigators and courts understand the digital evidence.

Q16. Explain key components in Report Generation in Digital Forensics.

Answer:

A digital forensic report must be accurate, clear, complete and based on documented evidence.

Key components:

1. Case identification: Case number, title, investigator and date.
2. Executive summary: Short overview of the investigation and main findings.
3. Objective and scope: Explains what was examined and what was outside the scope.
4. Legal authorization: Records the authority or permission for examination.
5. Evidence inventory: Lists each device, media item and evidence ID.
6. Chain of custody: Documents collection, handling, transfers and storage.
7. Tools and methodology: Names tools, versions, settings and procedures.
8. Integrity verification: Includes hash algorithms and recorded hash values.
9. Examination results: Lists extracted files, logs, metadata and other artefacts.
10. Analysis and timeline: Explains how evidence relates to the incident.
11. Findings and conclusion: States supported results without unsupported assumptions.
12. Limitations: Explains technical or evidential constraints.
13. Appendices: Provides screenshots, logs, tables and supporting material.
14. Examiner certification: Includes examiner identity, signature and report date.

Conclusion: These components make the report understandable,

transparent and suitable for independent review.

Q17. Explain the process of Examination and Analysis of Evidence.

Answer:

Examination and analysis are two connected but different stages of digital forensics.

Examination:

- Verifies evidence labels, seals and hash values.
- Creates or uses a verified forensic working copy.
- Identifies partitions and file systems.
- Extracts files, logs, browser history and metadata.
- Searches for relevant and deleted information.
- Records all actions and results.

Analysis:

- Interprets the extracted information.
- Correlates records from different devices and sources.
- Builds a timeline of relevant events.
- Identifies possible user actions and attack methods.
- Checks findings against independent evidence.
- Records conclusions and limitations.

Flow:

Preserved Evidence
↓
Verify Integrity
↓
Examination – Extract Data
↓
Analysis – Interpret Data
↓
Correlate & Build Timeline
↓
Document Findings
↓
Final Report

Example:

During an investigation, a forensic examiner extracts browser history, login logs and deleted files from a forensic image. The analyst compares timestamps and records to determine whether the evidence supports unauthorized access.

Conclusion: Examination finds and extracts relevant data, while analysis interprets it. Both stages must be documented and performed while maintaining evidence integrity.